

Year 10 Science — Flashcard Content

Course: Year 10 Science (Victorian Curriculum F-10, VCAA) **Source:** Zoe's textbook chapters — Genetics, The Periodic Table, Energy, Motion **Purpose:** Master content for flashcards. Format is deliberately simple so it can later be turned into PowerPoint, a web app, Anki, printed cards, etc.

How to read this file

- Each card is a ****Q:**** (front) followed by ****A:**** (back).
 - Cards are grouped by chapter (##) and topic (###) — useful for making decks/sub-decks.
 - One fact per card, kept short for quick recall.
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Chapter 2 — Genetics

Victorian Curriculum: VCSSU119 — The transmission of heritable characteristics from one generation to the next involves DNA and genes.

DNA structure

Q: What does DNA stand for? **A:** Deoxyribonucleic acid.

Q: What is genetics? **A:** The study of inheritance — how traits are passed from parents to offspring.

Q: Where is DNA found in a cell? **A:** In the nucleus (also in mitochondria and chloroplasts).

Q: What is the overall shape of a DNA molecule? **A:** A double helix — a twisted, double-stranded “ladder”.

Q: What is the basic building block (subunit) of DNA called? **A:** A nucleotide.

Q: What three parts make up a nucleotide? **A:** A five-carbon sugar (deoxyribose), a phosphate group, and a nitrogenous base.

Q: Name the four nitrogenous bases in DNA. **A:** Adenine (A), Thymine (T), Guanine (G), Cytosine (C).

Q: State the complementary base-pairing rule for DNA. **A:** A always pairs with T; G always pairs with C.

Q: How many hydrogen bonds form between A-T and between G-C? **A:** A-T has 2 hydrogen bonds; G-C has 3 hydrogen bonds.

Q: Which bases are purines and which are pyrimidines? **A:** Purines (double ring): adenine and guanine. Pyrimidines (single ring): cytosine and thymine.

Q: What forms the “backbone” (sides) of the DNA ladder? **A:** The alternating sugar and phosphate groups (the sugar-phosphate backbone).

Q: What holds the two strands of DNA together? **A:** Hydrogen bonds between the complementary nitrogenous bases.

Chromosomes, genes and proteins

Q: What is a chromosome? **A:** A structure of tightly wound DNA (coiled around proteins called histones).

Q: What is a gene? **A:** A length (section) of DNA/chromosome that holds the instructions for making a protein — the unit of inheritance.

Q: Describe the relationship between DNA, genes and chromosomes. **A:** DNA is wound into chromosomes; genes are sections of the DNA on a chromosome.

Q: What is a triplet? **A:** A group of three DNA bases that codes for one amino acid.

Q: What are amino acids? **A:** The subunits (building blocks) of proteins.

Q: Why are proteins important? **A:** They build and repair tissue, speed up reactions (enzymes), transport substances, fight invaders, and move muscles.

Q: What are the two processes needed to make a protein from a gene? **A:** Transcription and translation.

Q: What happens during transcription and where? **A:** In the nucleus, the DNA code is copied into messenger RNA (mRNA).

Q: What happens during translation and where? **A:** At the ribosomes, the mRNA code is decoded to build a chain of amino acids (a protein).

Q: List three differences between DNA and RNA. **A:** DNA is double-stranded, has deoxyribose sugar, and uses thymine; RNA is single-stranded, has ribose sugar, and uses uracil instead of thymine.

Q: What is a codon? **A:** A group of three bases on mRNA that specifies one amino acid.

Q: Which codon is the START codon, and what amino acid does it code for? **A:** AUG — it codes for methionine (Met).

Cell division and reproduction

Q: What is a haploid cell? (symbol) **A:** A cell with only one set of chromosomes (n). Example: gametes (sperm, egg).

Q: What is a diploid cell? (symbol) **A:** A cell with two sets of chromosomes ($2n$). Example: somatic (body) cells and the zygote.

Q: What is the haploid (n) and diploid ($2n$) number in humans? **A:** Haploid = 23; diploid = 46.

Q: What is mitosis? **A:** Cell division producing two genetically identical diploid daughter cells, used for growth and repair.

Q: What is meiosis? **A:** Cell division in the gonads producing four genetically unique haploid gametes (for sexual reproduction).

Q: Why is meiosis called a “reduction division”? **A:** Because the daughter cells (gametes) have half the genetic material of the parent cell.

Q: What are gametes and where are they made? **A:** Sex cells (sperm and ova); made in the gonads — testes in males, ovaries in females.

Q: What is fertilisation? **A:** The fusing of an egg and sperm to form a zygote.

Q: What is a zygote? **A:** A fertilised egg (diploid) formed when sperm and egg combine.

Q: What is crossing over and when does it occur? **A:** When homologous chromosomes swap genetic material during prophase I of meiosis — it increases variation.

Q: List two key differences between mitosis and meiosis. **A:** Mitosis makes 2 identical diploid cells; meiosis makes 4 unique haploid cells. Meiosis reshuffles genes by crossing over; mitosis does not.

Q: What is a karyotype? **A:** A pictorial representation of an organism's chromosomes arranged in pairs by size and pattern.

Q: What are autosomes? **A:** Chromosome pairs 1–22 (the non-sex chromosomes).

Q: What are the sex chromosomes in humans? **A:** The 23rd pair: XX = female, XY = male.

Q: Which parent determines the sex of a child, and why? **A:** The father — he can pass on either an X or a Y chromosome; the mother only passes X.

Patterns of inheritance

Q: What is an allele? **A:** A different form (version) of the same gene.

Q: What is a genotype? **A:** The combination of alleles an organism has for a gene (e.g. Tt).

Q: What is a phenotype? **A:** The physical appearance of a trait, resulting from genotype interacting with the environment.

Q: What does homozygous mean? **A:** Having two identical alleles for a trait (e.g. TT or tt).

Q: What does heterozygous mean? **A:** Having two different alleles for a trait (e.g. Tt).

Q: What is a dominant trait? **A:** A trait expressed when only one copy of its allele is present (RR or Rr).

Q: What is a recessive trait? **A:** A trait expressed only when two copies of its allele are present (rr).

Q: What is a locus? **A:** The location (position) of a gene on a chromosome.

Q: Who is known as the “father of genetics”? **A:** Gregor Mendel (used pea plants).

Q: What is a Punnett square used for? **A:** A grid to predict the genotypes and phenotypes of offspring from a cross.

Q: Cross $Tt \times Tt$ — what are the genotype and phenotype ratios? **A:** Genotype 1 TT : 2 Tt : 1 tt; phenotype 3 dominant : 1 recessive.

Q: What is a test cross? **A:** Crossing an organism showing a dominant trait with a homozygous recessive individual to find if it is homozygous or heterozygous.

Q: What is codominance? Give an example. **A:** When both alleles are expressed equally in the heterozygote. Example: AB blood type; black-and-white speckled chickens.

Q: What is incomplete dominance? Give an example. **A:** When neither allele is dominant, so the phenotype is a blend. Example: red $\times$ white snapdragons giving pink flowers.

Q: Why are males more affected by X-linked recessive disorders than females? **A:** Males have only one X chromosome, so one recessive allele is enough to show the trait; females need two.

Q: Name two X-linked recessive conditions. **A:** Haemophilia and red-green colour blindness (also Duchenne muscular dystrophy).

Q: What is a pedigree? **A:** A chart (like a family tree) used to study how a trait is inherited over generations.

Mutations and biotechnology

Q: What is a mutation? **A:** A change in the genetic code (DNA) of a cell.

Q: What is a mutagen? **A:** An agent that causes mutations (e.g. radiation, UV, certain chemicals).

Q: Difference between a spontaneous and an induced mutation? **A:** Spontaneous = occurs naturally; induced = caused by exposure to a mutagen.

Q: Difference between a germline and a somatic mutation? **A:** Germline mutations occur in gametes and can be inherited; somatic mutations occur in body cells and are not passed on.

Q: Name the three possible effects of a mutation. **A:** Neutral (no effect), deleterious (harmful), or beneficial (advantageous).

Q: What is a point mutation? **A:** A mutation that changes a single nucleotide.

Q: Name the four types of point mutation. **A:** Substitution, insertion, deletion, and inversion.

Q: Which point mutations cause a frameshift, and why? **A:** Insertion and deletion — they shift the reading of all following triplets/codons.

Q: What is aneuploidy? **A:** When an individual has one more or one fewer chromosome than normal.

Q: What is non-disjunction? **A:** When chromosomes fail to separate correctly during meiosis, giving gametes the wrong number of chromosomes.

Q: What is trisomy 21 / Down syndrome? **A:** Having three copies of chromosome 21 instead of two (an example of aneuploidy).

Q: What is a genetically modified organism (GMO)? **A:** An organism whose genome has been altered by humans.

Q: What is a transgenic organism? **A:** An organism that contains a “foreign” gene transferred from another species.

Q: What is gene therapy? **A:** Introducing a functional copy of a gene into an organism to produce a missing protein and treat an inherited disorder.

Chapter 4 — The Periodic Table

Victorian Curriculum: VCSSU123 — The atomic structure and properties of elements are used to organise them in the periodic table.

Development of the periodic table

Q: What is an element? **A:** A species of atoms that all have the same number of protons.

Q: What were Döbereiner's "triads"? **A:** Groups of three elements with similar properties, where the middle element's atomic mass was the average of the other two.

Q: Who is considered the creator of the first modern periodic table? **A:** Dmitri Mendeleev (1869).

Q: How did Mendeleev arrange his elements? **A:** In order of increasing atomic mass, grouping elements with similar properties.

Q: What was clever about Mendeleev leaving gaps in his table? **A:** He predicted that undiscovered elements would fill them, and predicted their properties (e.g. "eka-silicon").

Q: Which real element matched Mendeleev's "eka-silicon" prediction? **A:** Germanium (discovered 1886).

Q: How is the modern periodic table ordered (and how is this different to Mendeleev)? **A:** By increasing atomic number (number of protons), not atomic mass.

Structure of the periodic table

Q: What are groups in the periodic table? **A:** The vertical columns (there are 18).

Q: What are periods in the periodic table? **A:** The horizontal rows (there are 7).

Q: Why do elements in the same group have similar properties? **A:** They have the same number of valence (outer-shell) electrons, so they react similarly.

Q: What is the atomic number of an element? **A:** The number of protons in its atoms.

Q: List four properties of metals. **A:** Shiny, malleable, ductile, good conductors of heat/electricity (also high melting points, dense).

Q: List four properties of non-metals. **A:** Dull, brittle, poor conductors, low melting/boiling points.

Q: What is the only metal that is liquid at room temperature? **A:** Mercury (Hg).

Q: What is the only non-metal that is liquid at room temperature? **A:** Bromine (Br).

Q: What is a metalloid? **A:** An element near the "step ladder" border that shares properties of both metals and non-metals (e.g. silicon, boron).

Q: Where are the transition metals located? **A:** The block of metals in groups 3-12 (periods 4-7); e.g. iron, copper, silver, gold.

Electronic configuration

Q: What is an electron shell? **A:** A region around the nucleus that houses electrons.

Q: What is the maximum number of electrons in shells 1, 2, 3 and 4? **A:** Shell 1 = 2, shell 2 = 8, shell 3 = 18, shell 4 = 32.

Q: In what order do electrons fill the shells? **A:** From the lowest energy level first — the shell closest to the nucleus.

Q: What is electronic configuration? **A:** The arrangement of an atom's electrons in shells (e.g. magnesium = 2.8.2).

Q: What is the "ground state" of an atom? **A:** When its electrons are at their lowest possible energy levels.

Q: What are valence electrons? **A:** The electrons in the outermost shell of an atom.

Q: Why are valence electrons important? **A:** They are the most involved in chemical reactions and determine an element's properties and group.

Q: How do you find an element's group and period from its electron configuration? **A:** The last number = number of valence electrons (relates to group); the number of shells (numbers listed) = the period.

Q: What happens when electrons are "excited"? **A:** They gain energy and jump to a higher shell; when they fall back to ground state they release the energy as light.

Q: How can flame tests / emission spectra identify elements? **A:** Each element emits a characteristic colour/pattern of light when its excited electrons return to ground state.

Ions and bonding

Q: What is an ion? **A:** A charged atom, formed by losing or gaining electrons.

Q: What is a cation, and which elements form them? **A:** A positively charged ion (lost electrons) — formed by metals.

Q: What is an anion, and which elements form them? **A:** A negatively charged ion (gained electrons) — formed by non-metals.

Q: What is an ionic bond? **A:** A strong bond between a cation and an anion, formed by transferring (donating) electrons (metal + non-metal).

Q: What is a covalent bond? **A:** A strong bond formed when two non-metals share electrons to form a molecule.

Q: Why do ionic compounds have high melting points? **A:** The electrostatic attraction in the ionic bond/lattice is very strong (e.g. NaCl melts at 801 °C).

Special groups

Q: What is group 1 called and how many valence electrons do they have? **A:** The alkali metals — 1 valence electron.

Q: What charge ion do alkali metals form, and how reactive are they? **A:** +1 ions; highly reactive (reactivity increases down the group).

Q: What do alkali metals produce when they react with water? **A:** Hydrogen gas and an alkaline solution (a metal hydroxide).

Q: What is group 2 called and what charge ion do they form? **A:** The alkaline earth metals — they form +2 ions (less reactive than group 1).

Q: What is group 17 called, how many valence electrons, and what charge ion? **A:** The halogens — 7 valence electrons, forming -1 ions.

Q: What happens to halogen reactivity down the group? **A:** It decreases (opposite to group 1) — it gets harder to gain an electron.

Q: Why are halogens described as “diatomic”? **A:** They form molecules of two atoms (e.g. Cl_2 , Br_2).

Q: What is group 18 called and why are they unreactive? **A:** The noble gases — they have a full outer shell, so they don't need to gain/lose electrons (also called inert; they don't form ions).

Reactivity and extracting metals

Q: What is the activity series? **A:** A list of metals ordered by reactivity, from highest to lowest.

Q: Order these from most to least reactive: gold, potassium, zinc, copper. **A:** Potassium > zinc > copper > gold.

Q: What is a displacement reaction? **A:** A reaction where a more reactive metal removes (displaces) a less reactive metal from its compound.

Q: What is an ore? **A:** A naturally occurring compound (impure form) from which a metal can be extracted.

Q: What are native metals? Give examples. **A:** Unreactive metals found in pure form in nature (e.g. gold, silver).

Q: How are the most reactive metals (e.g. sodium, potassium) extracted? **A:** By electrolysis (carbon is not reactive enough).

Q: How are mid-reactivity metals (e.g. iron, zinc) extracted? **A:** By reaction with carbon, which removes the oxygen from the ore.

Q: Why is it harder to extract more reactive metals from their ores? **A:** The more easily a metal loses electrons, the harder it is to make it take them back (reverse the reaction).

Chapter 8 — Energy

Victorian Curriculum: VCSSU132 — Energy flow in Earth's atmosphere can be explained by the processes of heat transfer; the Law of Conservation of Energy.

Forms of energy and work

Q: What is energy? **A:** The capacity to cause change — defined as the ability to do work.

Q: What unit is energy measured in? **A:** Joules (J).

Q: What is work (in physics)? **A:** The energy transferred when an applied force causes an object to move some distance.

Q: State the equation for work and the units. **A:** $W = Fs$, where W is work (J), F is force (N), and s is displacement (m).

Q: When is no work done on an object? **A:** When the object does not move (no displacement) in the direction of the force.

Q: What are the two main types of energy that all forms fall under? **A:** Kinetic energy and potential energy.

Q: What is kinetic energy? **A:** The energy an object has due to its motion.

Q: State the equation for kinetic energy. **A:** $K = \frac{1}{2} m v^2$ (mass in kg, speed in m/s).

Q: In the KE equation, what has a bigger effect — doubling mass or doubling speed? **A:** Doubling speed (KE goes up $\times 4$, because speed is squared); doubling mass only doubles KE.

Q: What is potential energy? **A:** Energy stored in an object due to its position or condition.

Q: What is gravitational potential energy (GPE)? **A:** Energy stored in an object due to its height above the surface of Earth.

Q: State the equation for gravitational potential energy. **A:** $U_g = m \times g \times h$ (mass in kg, $g = 9.8 \text{ m/s}^2$ on Earth, height in m).

Q: What is elastic potential energy? **A:** Energy stored in an object due to its shape — when stretched or compressed.

Q: Name some other forms of potential energy. **A:** Chemical, nuclear, electrical (charge) potential energy.

Thermal energy, temperature and heat

Q: What is thermal energy? **A:** The total kinetic energy of all the particles (atoms) that make up an object or system.

Q: What is temperature? **A:** A measure of the average kinetic energy of the particles in an object or system.

Q: What is heat? **A:** Thermal energy transferred between objects due to a temperature difference.

Q: Can a large object have more thermal energy than a hotter small object? Explain. **A:** Yes — thermal energy depends on the number of particles too, so a big cooler object can hold more total thermal energy than a small hot one.

Q: What temperature is absolute zero, in °C and K? **A:** $0 \text{ K} = -273.15 \text{ }^\circ\text{C}$ (the point where particles stop moving).

Q: What is thermal equilibrium? **A:** When two bodies are at the same temperature, so there is no net transfer of thermal energy between them.

Q: State the law of conservation of energy. **A:** Energy cannot be created or destroyed; it can only be transformed into other forms or transferred to other objects.

Q: State the first law of thermodynamics. **A:** Energy cannot be created or destroyed; it can only be transformed or transferred (so the total energy of an isolated system is constant).

Heat transfer

Q: What is conduction? **A:** Transfer of thermal energy through/between substances via collisions between particles.

Q: Why are solids and liquids better conductors than gases? **A:** Their particles are more densely packed, so there are more collisions and more energy transfer.

Q: What are conductors and insulators? **A:** Conductors transfer heat easily (e.g. metals); insulators are poor conductors of heat.

Q: What is convection? **A:** Transfer of thermal energy by the movement of particles in a fluid (gas or liquid) — forming convection currents.

Q: Why do convection currents form? **A:** Warm fluid is less dense and rises; cool fluid is denser and sinks, creating a circulating current.

Q: What is radiation (heat transfer)? **A:** Transfer of thermal energy without any particles of matter — e.g. infrared electromagnetic radiation.

Q: How does energy from the Sun reach Earth, and why? **A:** By radiation — there are no particles in the vacuum of space for conduction or convection.

Efficiency and energy systems

Q: Why is no energy conversion 100% efficient? **A:** Some energy is always lost from the system, usually as “wasted” heat and sound.

Q: What is useful energy? **A:** Energy that can be used for a specific (intended) purpose.

Q: State the equation for efficiency. **A:** Efficiency = (useful energy output ÷ energy input) × 100%.

Q: A light globe takes in 75 J and produces 12 J of light. What is its efficiency? **A:** $(12 \div 75) \times 100 = 16\%$.

Q: What is a Sankey diagram? **A:** A flow chart that shows the flow of energy through a system, with arrow widths proportional to the amount of energy.

Q: What is a “system” in physics? **A:** A portion of the universe that is being analysed.

Q: Describe the energy conversions in a swinging pendulum. **A:** Gravitational potential energy $\rightleftharpoons$ kinetic energy, repeatedly; energy is gradually lost as heat/sound (mainly from air resistance) until it stops.

Q: How do crumple zones reduce injury in a car crash? **A:** They crumple to absorb energy and increase the time/distance of the collision, reducing the force on occupants ($W = Fs$, so larger s means smaller F).

Q: How do seatbelts and airbags reduce injury? **A:** They increase the area, time and distance over which the stopping force acts, reducing the force on the body.

Q: How does a refrigerator move heat from cold food to the warmer room? **A:** By doing work — electrical energy powers a pump that circulates a coolant, which absorbs heat inside and releases it outside.

Chapter 9 — Motion

Victorian Curriculum: VCSSU133 — The motion of objects involves the interaction of forces and the exchange of energy, and can be described and predicted using the laws of physics.

Describing motion

Q: What is distance? **A:** The total length of the path travelled (symbol d , measured in metres). It has size only.

Q: What is displacement? **A:** How far and in what direction an object is from its starting point. It has size and direction.

Q: What is the key difference between distance and displacement? **A:** Distance is the total path length; displacement also includes direction and measures change in position (so it can be zero even after moving).

Q: What is speed? **A:** A measure of how fast something travels — the distance travelled per unit time.

Q: State the equation for average speed. **A:** $v_{av} = d \div t$ (distance $\div$ time).

Q: What units are commonly used for speed? **A:** Metres per second (m/s) and kilometres per hour (km/h).

Q: How do you convert m/s to km/h, and km/h to m/s? **A:** m/s $\rightarrow$ km/h: multiply by 3.6. km/h $\rightarrow$ m/s: divide by 3.6.

Q: Difference between average speed and instantaneous speed? **A:** Average speed is over a whole journey (A to B); instantaneous speed is the speed at a particular instant (e.g. a speedometer reading).

Q: What is velocity? **A:** The rate of change of displacement — speed in a particular direction (a vector).

Q: How is velocity different from speed? **A:** Velocity includes direction (uses displacement); speed does not (uses distance).

Q: What is acceleration? **A:** The rate at which an object changes its velocity (speed).

Q: State the equation and unit for acceleration. **A:** $a = (v_2 - v_1) \div t$; unit is m/s^2 (metres per second per second).

Q: What is deceleration? **A:** Negative acceleration — when an object slows down.

Q: What is constant speed? **A:** When an object travels at the same speed and is not accelerating or decelerating (acceleration = 0).

Graphing motion

Q: On a distance–time graph, what does the gradient represent? **A:** The speed of the object (rise $\div$ run).

Q: On a distance–time graph, what does a horizontal line mean? **A:** The object is stationary (gradient = 0, so speed = 0).

Q: On a distance–time graph, what does a steeper line mean? **A:** A greater speed.

Q: What does a displacement–time graph show that a distance–time graph doesn't? **A:** Direction — whether the object is moving away from or back towards its starting point (origin).

Q: On a speed–time graph, what does the gradient represent? **A:** The acceleration of the object.

Q: On a speed-time graph, what does the area under the line represent? **A:** The distance travelled.

Q: On a speed-time graph, what does a horizontal line mean? **A:** Constant speed (zero acceleration).

Q: On a speed-time graph, what do positive and negative gradients mean? **A:** Positive gradient = accelerating (speeding up); negative gradient = decelerating (slowing down).

Forces and Newton's laws

Q: What is a force? **A:** A push or pull — any interaction that, if unopposed, will change the motion of an object (measured in newtons, N).

Q: What is net force? **A:** The sum of all the forces acting on an object.

Q: What does it mean if the net force on an object is zero? **A:** The forces are balanced — the object stays at rest or keeps moving at constant speed in a straight line.

Q: State Newton's first law of motion. **A:** An object will stay at rest, or keep moving at constant speed in a straight line, unless acted on by an unbalanced (net) force.

Q: What is inertia? **A:** The tendency of an object to resist a change in its motion (stay at rest or keep moving the same way).

Q: State Newton's second law of motion (in words and as a formula). **A:** The net force equals mass times acceleration: $F_{\text{net}} = ma$.

Q: Using $F = ma$, what happens to acceleration if you apply the same force to a more massive object? **A:** The acceleration is smaller (acceleration is inversely proportional to mass).

Q: State Newton's third law of motion. **A:** For every action there is an equal but opposite reaction force.

Q: In Newton's third law, what's true about the action-reaction pair? **A:** They are equal in size, opposite in direction, and act on different objects.

Q: What is mass? **A:** The quantity of matter in an object (measured in kilograms); it does not change with location.

Q: What is weight? **A:** The force of gravity acting on an object (measured in newtons): $W = mg$.

Q: State the equation for weight (force due to gravity). **A:** $F_g = m \times g$ (mass in kg $\times$ gravitational acceleration in m/s^2).

Q: What is the value of g (acceleration due to gravity) on Earth? **A:** 9.8 m/s^2 .

Q: Why does an object weigh less on the Moon than on Earth, even though its mass is the same? **A:** The Moon's gravitational field is weaker ($g = 1.6 \text{ m/s}^2$ vs 9.8 m/s^2), so the weight force is smaller, but mass is unchanged.